Review Questions

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Q1. What is the purpose of camera_matrix in the code? Does the camera matrix remain the same for all devices?

Ans. The camera_matrix contains the camera's intrinsic parameters, which help OpenCV understand how the 3D world is projected onto a 2D image. It includes values such as the focal lengths and the optical centre of the camera. These values are used to estimate the ArUco marker's position and orientation accurately.

The camera matrix does not necessarily remain the same for all devices. Different cameras may have different focal lengths, lenses and sensor characteristics. Therefore, for accurate pose estimation, the camera should ideally be calibrated, and the matrix should correspond to that particular camera and image resolution.

Q2. Why is the orientation coefficient initialized to 0 in the code?

Ans. An orientation vector initialized to zero represents an initial assumption that the object has no rotation about the three axes. However, this is only a starting value and does not necessarily represent the marker's actual orientation. In my code, the marker's orientation is estimated using cv2.solvePnP() and stored in rvec, which contains the rotation information along the three axes.

Q3. What is the purpose of if __name__ == "__main__": in Python? Will the code inside this block execute if __main__ is changed to another name?

Ans. The statement if __name__ == "__main__": checks whether a Python file is being run directly or imported into another Python file. When the file is run directly, Python sets __name__ to "__main__", so the code inside this block executes. If the file is imported as a module, the code inside the block does not execute automatically.

If "__main__" is changed to another name, such as "start", the condition will normally be false when the file is run directly, so the code inside the block will not execute. This check is useful because it lets us run the main program only when intended, while still allowing functions from the file to be imported and reused elsewhere.